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The Ax–Grothendieck Theorem

From Infinite Fields to Finite Fields

Arkapriyo Hore

Universität Münster

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The theorem

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Theorem (Ax–Grothendieck, affine form)

Let $\mathbb{K}$ be an algebraically closed field and let

$$P : \mathbb{K}^n \longrightarrow \mathbb{K}^n$$

be a polynomial map. If P is injective, then P is surjective.

We write

$$P = (P_1, \dots, P_n), \quad P_i \in \mathbb{K}[x_1, \dots, x_n].$$

Plan of the talk

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- 1 Motivation: why this looks like linear algebra.
- 2 Brief analytic overview over $\mathbb{C}$.
- 3 Main proof: finite fields + Nullstellensatz + spreading out.
- 4 A fixed-point theorem for finite p -groups.

Core philosophy:

reduce an infinite-field problem to a finite-field problem.

Motivation I: dimension one

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For $n = 1$, let

$$P : \mathbb{K} \longrightarrow \mathbb{K}$$

be a polynomial map over an algebraically closed field.

If P is injective, then P is nonconstant. Hence by algebraic closedness,

$$P(x) = a$$

has a solution for every $a \in \mathbb{K}$.

So in dimension one the theorem is essentially the Fundamental Theorem of Algebra.

Motivation II: linear algebra

For a linear map

$$T : \mathbb{K}^n \longrightarrow \mathbb{K}^n,$$

finite-dimensional linear algebra says

$$T \text{ injective} \iff T \text{ surjective.}$$

For polynomial maps, one implication fails:

$$\mathbb{C} \longrightarrow \mathbb{C}, \quad z \longmapsto z^2$$

is surjective but not injective.

Ax–Grothendieck says that the other implication survives:

$$P \text{ injective} \implies P \text{ surjective.}$$

Why varieties are not vector spaces

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A general algebraic variety is not naturally a vector space.

Suppose a singular curve C had a vector-space-like addition $\oplus$ by morphisms. Then translations

$$T_Q : C \rightarrow C, \quad P \mapsto P \oplus Q$$

would be isomorphisms.

If P_0 is singular, then $T_Q(P_0) = P_0 \oplus Q$ is singular for every Q . As Q varies, this would make every point singular.

But for a curve,

$$\text{Sing}(C) \subsetneq C.$$

So affine space is special: it has both algebraic and linear structure.

A complex-analytic route over $\mathbb{C}$

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There is a proof over $\mathbb{C}$ using complex analysis.

Lemma (Rosay)

Let $\Omega \subseteq \mathbb{C}^n$ be open and let

$$f : \Omega \rightarrow \mathbb{C}^n$$

be injective and holomorphic. Then

$$\det Df(z) \neq 0 \quad \forall z \in \Omega.$$

Thus an injective holomorphic map is locally biholomorphic, hence open.

Rudin uses this openness to prove Ax–Grothendieck over $\mathbb{C}$.

Nullstellensatz as a finite obstruction

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Let $\mathbb{K}$ be algebraically closed.

Theorem (Hilbert's Nullstellensatz, (algebraic form))

If $g_1, \dots, g_r, h \in \mathbb{K}[x_1, \dots, x_n]$ and h vanishes at every common zero of the g_i , then

$$h^N = \sum_{i=1}^r g_i Q_i$$

for some $N \geq 1$ and $Q_i \in \mathbb{K}[x_1, \dots, x_n]$.

No solution implies a finite algebraic obstruction.

Injectivity as a finite algebraic condition

Let

$$P = (P_1, \dots, P_n) : \mathbb{K}^n \rightarrow \mathbb{K}^n.$$

Injectivity says:

$$P(x) = P(y) \implies x = y.$$

Equivalently, for each j , the polynomial $x_j - y_j$ vanishes on the common zero set of

$$P_1(x) - P_1(y), \dots, P_n(x) - P_n(y).$$

By Nullstellensatz, for each j there are $N_j \geq 1$ and polynomials $Q_{i,j}$ such that

$$(x_j - y_j)^{N_j} = \sum_{i=1}^n (P_i(x) - P_i(y)) Q_{i,j}(x, y).$$

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Failure of surjectivity as a finite algebraic condition

If P is not surjective, choose

$$z_0 = (z_1, \dots, z_n) \notin P(\mathbb{K}^n).$$

Then the equations

$$P_i(x) - z_i = 0, \quad i = 1, \dots, n,$$

have no common solution.

By Nullstellensatz,

$$(P_1(x) - z_1, \dots, P_n(x) - z_n) = \mathbb{K}[x_1, \dots, x_n].$$

So there are polynomials $R_1, \dots, R_n$ such that

$$\sum_{i=1}^n R_i(x) (P_i(x) - z_i) = 1.$$

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Finite fields: the elementary core

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Let $\mathbb{F}$ be a finite field.

Lemma

If

$$P : \mathbb{F}^n \rightarrow \mathbb{F}^n$$

is injective, then P is bijective.

This does not use polynomials.

It is just the elementary fact:

an injective self-map of a finite set is surjective.

The whole proof tries to reduce to this statement.

Proof(Ax-Grothendieck)

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Assume for contradiction that P is injective but not surjective.

From P being injective and not surjective, we collect all coefficients occurring in

$$P_i, \quad Q_{i,j}, \quad R_i, \quad z_0.$$

Call this finite set

$$C = \{\alpha_1, \dots, \alpha_r\} \subset \mathbb{K}.$$

The key point:

the whole contradiction uses only finitely many coefficients.

This is what makes reduction to finite fields possible.

Spreading out

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Let A be the ring generated by C over the prime subring of $\mathbb{K}$:

$$A = \begin{cases} \mathbb{Z}[\alpha_1, \dots, \alpha_r], & \text{char}(\mathbb{K}) = 0, \\ \mathbb{F}_p[\alpha_1, \dots, \alpha_r], & \text{char}(\mathbb{K}) = p > 0. \end{cases}$$

Then all our polynomial identities are defined over A .

Choose a maximal ideal

$$\mathfrak{m} \subset A, \quad \kappa = A/\mathfrak{m}.$$

Reducing coefficients modulo $\mathfrak{m}$ gives

$$\overline{P} : \kappa^n \rightarrow \kappa^n.$$

Why the reduced map is still bad

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The identities survive reduction modulo $\mathfrak{m}$.

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The injectivity after reduction becomes

$$(x_j - y_j)^{N_j} = \sum_i (\overline{P}_i(x) - \overline{P}_i(y)) \overline{Q}_{i,j}(x, y).$$

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Thus $\overline{P}$ is injective.

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The non-surjectivity after reduction becomes

$$\sum_i \overline{R}_i(x) (\overline{P}_i(x) - \overline{z}_i) = 1.$$

Thus $\overline{z}_0$ is not in the image of $\overline{P}$.

So $\overline{P}$ is injective but not surjective.

Why is κ finite? Positive characteristic

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If

$$A = \mathbb{F}_p[\alpha_1, \dots, \alpha_r],$$

then

$$\kappa = A/\mathfrak{m} = \mathbb{F}_p[\bar{\alpha}_1, \dots, \bar{\alpha}_r]$$

is a field finitely generated as an $\mathbb{F}_p$ -algebra.

By Zariski's Lemma,

$$[\kappa : \mathbb{F}_p] < \infty.$$

Hence

$$|\kappa| = p^{[\kappa : \mathbb{F}_p]} < \infty.$$

So κ is a finite field.

Why is κ finite? Characteristic zero

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Now suppose

$$A = \mathbb{Z}[\alpha_1, \dots, \alpha_r].$$

We must show $\kappa = A/\mathfrak{m}$ is finite.

First prove

$$\text{char } \kappa \neq 0.$$

If $\text{char } \kappa = 0$, then

$$\kappa = \mathbb{Q}[\bar{\alpha}_1, \dots, \bar{\alpha}_r].$$

By Zariski's Lemma,

$$[\kappa : \mathbb{Q}] < \infty.$$

So the $\bar{\alpha}_i$ are algebraic over $\mathbb{Q}$.

The denominator obstruction

Clear finitely many denominators. Then for some $N \neq 0$,

$$B = \mathbb{Z} \left[\frac{1}{N} \right]$$

has the property that all $\bar{\alpha}_i$ are integral over B .

Hence

$$\kappa = B[\bar{\alpha}_1, \dots, \bar{\alpha}_r]$$

is integral over B .

But κ is a field. If a field is integral over B , then B is a field.

Contradiction: $\mathbb{Z}[1/N]$ is not a field; choose a prime $q \nmid N$, then

$$\frac{1}{q} \notin \mathbb{Z} \left[\frac{1}{N} \right].$$

Finishing the proof

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Thus in the characteristic-zero case too,

$$\text{char } \kappa = p > 0.$$

Then since $\mathfrak{m} \cap \mathbb{Z} = (p)$, we have

$$\kappa = \mathbb{F}_p[\bar{\alpha}_1, \dots, \bar{\alpha}_r]$$

is a field finitely generated over $\mathbb{F}_p$, hence finite by Zariski's Lemma.

So in all cases, κ is finite.

But we constructed

$$\bar{P} : \kappa^n \rightarrow \kappa^n$$

injective and not surjective.

This is impossible for a finite set.

Therefore P is surjective.

A Noetherian-module alternative

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Remark

Suppose, as before, that

$$\text{char}(\kappa) = 0.$$

We had obtained

$$\kappa = B[\alpha_1, \dots, \alpha_r], \quad B = \mathbb{Z}\left[\frac{1}{N}\right],$$

with each α_i integral over B . Hence κ is a finitely generated B -module.

Since B is Noetherian, κ is a Noetherian B -module.

A Noetherian-module alternative

But

$$\mathbb{Q} \subseteq \kappa$$

is a B -submodule, so $\mathbb{Q}$ would also be Noetherian as a B -module.

Choose distinct primes

$$q_1, q_2, q_3, \dots \nmid N.$$

Then we obtain a strictly increasing chain

$$B \subsetneq B\left[\frac{1}{q_1}\right] \subsetneq B\left[\frac{1}{q_1}, \frac{1}{q_2}\right] \subsetneq B\left[\frac{1}{q_1}, \frac{1}{q_2}, \frac{1}{q_3}\right] \subsetneq \dots$$

of B -submodules of $\mathbb{Q}$.

Hence $\mathbb{Q}$ is not Noetherian as a B -module, a contradiction.

$\text{char}(\kappa) \neq 0.$

Same philosophy: a fixed-point theorem

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Theorem

Let G be a finite p -group acting algebraically on $\mathbb{A}_k^n$, where k is algebraically closed and

$$\text{char } k \neq p.$$

Then the action has a fixed point.

Over a finite field $\mathbb{F}_{\ell^m}$ with $\ell \neq p$,

$$|\mathbb{A}^n(\mathbb{F}_{\ell^m})| = \ell^{mn},$$

which is not divisible by p .

A p -group orbit has size a power of p .

So at least one orbit has size 1.

Reducing the fixed-point theorem to finite fields

For a general algebraically closed k , assume there is no fixed point.

The fixed point equations are

$$x_i - P_{g,i}(x) = 0, \quad g \in G, \quad i = 1, \dots, n.$$

No solution means Nullstellensatz gives:

$$\sum_{g \in G} \sum_i (x_i - P_{g,i}(x)) Q_{g,i}(x) = 1.$$

Spread out the finitely many coefficients, reduce modulo a maximal ideal to a finite field of characteristic different from p , (in characteristic 0 this is accomplished by adjoining $1/p$ to the $\mathbb{Z}$ -algebra) and get a contradiction with the finite-field orbit-counting argument.

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Takeaway

Theorem (Corollary)

Let σ be an automorphism of $\mathbb{C}^n$, viewed as an affine variety. If $\sigma^2 = \text{id}$, then σ has a fixed point.

All these proofs are driven by three ideas:

- 1 Nullstellensatz turns geometric failure into finite algebraic data.
- 2 Spreading out puts this data over a finitely generated ring.
- 3 Closed-point reduction sends the problem to finite fields.

Then the theorems reduce to:

injective self-map of a finite set $\Rightarrow$ surjective.

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Thank You!

For your attention.

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